

LAUNCESTON TI TREE BEND, Australia

WMO: 949690

Lat: 41.4194S

Lon: 147.1222E

Elev: 16

StdP: 14.69

Time Zone: 10.00 (AUS)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.6	32.4	28.2	22.2	41.3	29.8	23.9	40.9	24.6	54.9	21.4	53.1	1.3	100	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	82.7	65.9	79.5	64.8	76.7	63.2	68.3	78.0	66.8	75.7	65.3	73.6	11.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	93.5	71.5	63.6	88.1	70.6	61.9	82.9	69.4	32.6	77.9	31.4	75.7	30.2	73.8	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.4	16.3	27.9	89.8	1.6	3.9	26.7	92.6	25.8	94.9	24.9	97.1	23.7	100.0		
(5)			27.8	71.8	2.2	1.8	26.2	73.1	25.0	74.2	23.7	75.2	22.2	76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	65.1	65.1	61.8	55.7	50.8	46.9	46.3	47.6	50.4	54.0	58.4	61.6	(6)
(7)		DBStd	8.04	5.03	4.67	5.12	4.57	4.45	4.17	3.71	3.87	3.68	4.28	4.68	4.69	(7)
(8)		HDD50	438	0	0	0	8	46	111	126	96	39	11	1	0	(8)
(9)		HDD65	3732	60	51	127	280	441	543	581	539	438	341	204	127	(9)
(10)		CDD50	2356	469	424	367	179	70	17	10	22	51	135	253	359	(10)
(11)		CDD65	175	64	55	28	1	0	0	0	0	0	6	6	21	(11)
(12)		CDH74	1285	507	344	181	4	0	0	0	0	0	6	67	176	(12)
(13)		CDH80	243	121	55	31	0	0	0	0	0	0	6	6	30	(13)
(14)	Wind	WSAvg	6.7	8.1	7.3	6.6	5.6	5.5	4.6	5.4	6.1	6.8	7.6	8.0	8.2	(14)
(15)	Precipitation	PrecAvg	26.3	2.0	1.1	1.3	2.1	2.6	2.7	3.1	3.2	2.6	2.2	1.9	1.8	(15)
(16)		PrecMax	30.9	5.5	4.2	4.2	4.9	5.5	7.2	6.2	6.4	4.6	4.8	3.3	5.7	(16)
(17)		PrecMin	19.4	0.4	0.0	0.1	0.7	0.3	0.8	1.5	0.6	0.3	0.7	0.2	0.2	(17)
(18)		PrecStd	3.3	1.3	0.9	1.1	1.1	1.4	1.3	1.2	1.7	1.1	1.1	0.7	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.0	85.8	84.2	73.8	65.9	61.1	59.0	61.8	67.0	73.8	80.4	84.1	(19)
MCWB			68.8	67.1	66.3	62.2	57.4	54.9	52.3	53.9	56.2	59.5	63.7	65.3	(20)	
2%		DB	83.0	81.3	78.6	70.1	63.1	58.6	56.9	59.1	63.3	69.0	75.3	78.6	(21)	
		MCWB	65.9	66.2	64.4	60.1	56.0	53.3	51.3	52.5	54.2	57.6	62.3	63.5	(22)	
5%		DB	79.3	78.1	75.1	67.5	61.1	56.8	55.3	57.1	61.0	66.2	71.6	75.1	(23)	
		MCWB	64.8	64.3	63.1	58.4	55.1	52.3	50.7	50.8	52.8	56.4	60.1	61.1	(24)	
10%		DB	75.9	75.5	71.9	65.0	59.3	55.1	53.9	55.4	59.0	63.7	68.4	72.1	(25)	
		MCWB	62.8	62.9	61.2	57.1	54.2	51.4	50.0	50.1	51.9	54.7	58.1	60.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	70.1	68.6	64.4	59.7	56.9	54.4	55.9	58.3	62.0	66.4	68.2	(27)
MCD B			84.0	79.4	78.7	70.1	62.9	59.2	57.0	59.5	64.0	68.8	74.5	76.4	(28)	
2%		WB	68.5	68.1	66.7	62.0	57.9	54.6	52.9	53.9	56.1	59.7	63.7	66.1	(29)	
		MCD B	78.5	77.4	75.3	67.3	61.1	57.2	55.1	57.2	61.1	66.3	72.3	74.7	(30)	
5%		WB	66.6	66.5	64.8	60.5	56.6	53.2	51.8	52.5	54.5	57.7	61.7	64.0	(31)	
		MCD B	75.8	75.0	72.0	65.6	59.8	55.9	54.3	55.8	59.0	63.9	69.5	72.5	(32)	
10%		WB	64.7	64.9	63.0	58.7	55.2	51.8	50.6	51.2	52.9	56.0	59.7	61.9	(33)	
		MCD B	73.5	72.9	70.0	63.5	58.6	54.6	53.3	54.4	57.3	62.2	67.0	70.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.1	19.6	19.5	18.8	17.1	16.3	15.8	16.3	17.3	18.5	18.7	19.3	(35)
MCDBR			24.5	22.8	22.6	21.5	17.9	16.9	16.4	17.7	19.6	21.7	23.3	23.5	(36)	
MCWBR			10.7	10.3	10.6	12.2	11.7	12.0	11.5	11.6	11.8	12.0	12.0	11.1	(37)	
5% WB		MCDBR	21.3	20.1	19.6	18.4	14.5	14.5	13.6	14.2	17.3	19.1	20.5	20.7	(38)	
		MCWBR	11.3	10.6	10.6	11.9	10.5	11.0	10.6	10.2	11.5	11.7	11.9	11.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.349	0.342	0.331	0.321	0.309	0.299	0.297	0.315	0.335	0.341	0.344	0.342	(40)	
taud		2.480	2.520	2.545	2.548	2.559	2.575	2.586	2.532	2.463	2.455	2.471	2.489	(41)		
Ebn at Noon		310	304	294	274	256	249	258	272	287	301	310	314	(42)		
Edh at Noon		36	33	30	25	21	19	20	25	31	35	36	36	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2216	1868	1444	977	634	491	544	799	1182	1654	1985	2197	(44)	
(45)		RadStd	150	76	64	52	40	43	27	64	81	86	142	143	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		+0.58	N/A	N/A	+1.21	+0.68	+0.81	N/A	-180	+194	+25

Nomenclature: See separate page